

Physical-Direction-Assisted Contact-Field Registration for Compliant Docking with Anchored Capture

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Abstract—A biased target pose can prevent compliant docking of a complex interface. We propose physical-direction-assisted contact-field registration (PCF): offline contact poses constrain field values, while directional cues constrain field slopes. Aligning online contact poses to this field estimates the target. Registration-anchored capture (RAC) then permits bounded, wrench-driven corrections around the registered trajectory. A constrained optimization balances contact response against an elastic anchor shaped by registration sensitivity. We establish a discrete storage inequality for this reference update, including finite-rotation step reduction. On 36 simulated boundary-bias inputs, PCF reduces mean lateral/rotation error from 1.478 mm/1.984° for Symmetric CMR to 0.428 mm/0.469°. On 12 hardware development replays, errors decrease from 1.619 mm/1.114° for a five-start baseline to 0.897 mm/0.619°. Across ten exploration groups, the full method meets the 0.5 mm/0.5° hold tolerance in 129/130 target-frame boundary captures, compared with 102/130 for registered compliance and 76/130 for capture at the prior. It completes 9/10 captures of a separate camera-bias condition. Matched controls delimit these gains; new-method hardware capture remains untested.

Index Terms—Compliant manipulation, contact-based registration, force control, robotic assembly.

I. INTRODUCTION

Robotic docking of complex interfaces requires accurate alignment despite uncertainty in the perceived target pose. Mechanical lead-in features and compliant execution accommodate finite error, but a biased reference can still drive the male interface against an incorrect contact region. The resulting compliant equilibrium need not coincide with the assembled pose. Contact feedback is therefore useful both for identifying the target before capture and for adjusting the subsequent reference.

Contact-manifold registration (CMR) estimates a target by matching observed contact configurations to an offline contact set [1]. For the lotus-shaped interface considered here, exploration yields intermittent, trajectory-dependent coverage of a high-dimensional configuration boundary. Geometry-only local fits can be poorly determined when the configurations do not excite every tangent direction. We use physical direction cues to constrain the contact field fitted to the offline configurations.

The relationship between contact geometry, compliance, and jamming is well established [2], [3]. Recent estimation methods use differentiable contact features [4], manifold sampling [1], or learned projections onto a contact manifold [5]. Implicit contact representations also support learning contact dynamics [6] and joint shape/pose inference during planar pushing [7], while learned force models enable online

insertion control [8]. Our physical cue acts during offline field construction: a coordinate-consistent generalized contact direction supplies a soft derivative observation. Online registration then requires contact poses and a target prior, without estimating contact locations or fitting an online wrench-to-pose model. Under friction and multiple contacts, the cue is used as a soft constraint on the field slope.

Registration leaves a control-design question: how should capture use contact feedback while retaining the geometric information already obtained? Prior work has studied information-dependent intervention [9] and proximal optimization [10]. Here, the registered trajectory serves as an elastic anchor: contact can move the reference, but accumulated correction incurs a restoring penalty. The anchor shape comes from registration residual sensitivity mapped to the capture coordinates. Each reference update balances wrench-driven correction and departure from that anchor under cumulative and command-rate limits. The analysis concerns the sampled reference subsystem; the separation between virtual and realized admittance remains relevant to robot execution [11].

This letter develops two connected components:

- PCF uses physical direction cues to combine contact positions and slopes in a global field and local affine patches for robust target estimation from online contact poses;
- RAC transfers registration sensitivity to a constrained reference update with a discrete storage inequality preserved by finite-rotation step reduction.

Figure 1 summarizes the method. Evaluation separates paired registration accuracy from four-system docking completion, with representative capture and camera-bias-range experiments. Simulation includes ten exploration groups and matched four-system runs; hardware registration replay and prospective capture measurements are reported separately, with the latter currently pending.

II. PROBLEM FORMULATION

We consider docking a male interface into a fixed target whose perceived pose is biased. The objective is to reduce this bias using contact observations and then accommodate the remaining error during capture. The axial approach is regulated separately; estimation and reference adjustment focus on two lateral translations and three rotations.

Let $\{B\}$, $\{M\}$, and $\{F\}$ denote the robot base, male-interface, and target frames. A transform ${}^AT_B \in \text{SE}(3)$ maps coordinates from frame $\{B\}$ to frame $\{A\}$. The true target

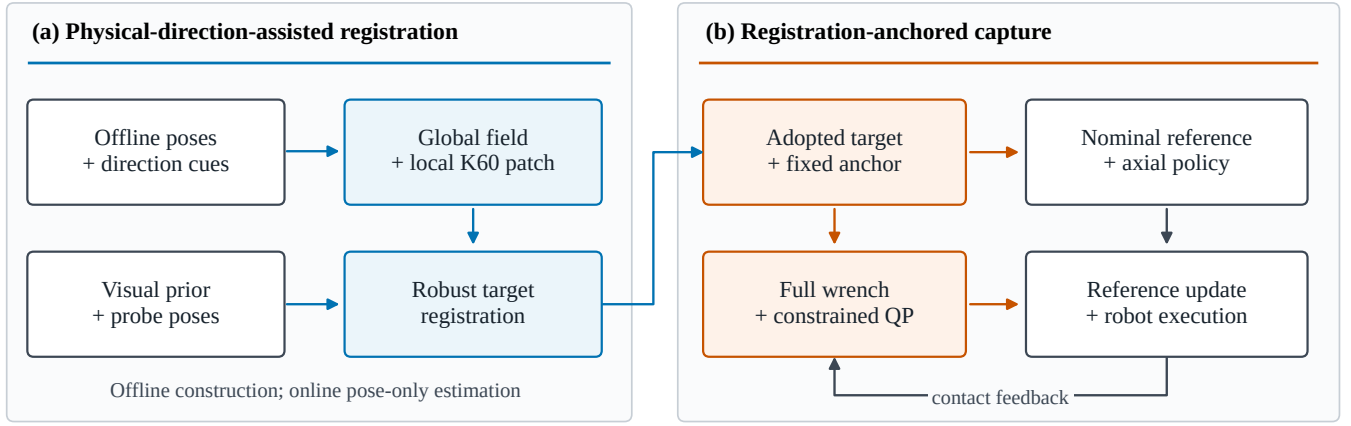


Fig. 1. Contact-field registration followed by anchored capture. (a) Offline contact configurations and physical direction cues constrain a global field and local K60 patches. Online contact poses and a visual prior produce a target estimate. (b) The adopted, limited target defines nominal capture; its frozen residual sensitivity shapes an elastic reference anchor. Full reference-frame wrench drives a constrained update. Reference execution and the axial policy are separate from the analyzed five-dimensional correction.

pose is $T_F = {}^B T_F$, and its visual prior is

$$\hat{T}_v = T_F \Delta T_v, \quad (1)$$

where ΔT_v is the unknown target-frame bias. Let $T_\star$ be the known male pose relative to the target at alignment. Hardware uses the calibrated aligned male frame as its target datum, so $T_\star = I_4$ (I_d denotes the d -dimensional identity); simulation uses the female CAD datum and retains a generally nonidentity $T_\star$. Under perfect tracking, commanding $\hat{T}_v T_\star$ therefore yields $\Delta T_v T_\star$ in the true target frame. Compliance alone need not remove this error.

Offline acquisition provides contact poses $\mathcal{M} = \{X_j\}_{j=1}^N$ relative to the known target, together with available physical direction cues. Online exploration records male poses $\mathcal{O} = \{O_i\}_{i=1}^{N_o}$ in the base frame. For a candidate target $\hat{T}$, their relative poses are

$$X_i(\hat{T}) = \hat{T}^{-1} O_i. \quad (2)$$

Registration seeks a target that makes these observations consistent with the offline contact set. We denote its raw estimate by $\tilde{T}$ and the estimate adopted after correction limits by $\hat{T}_r$. The latter defines the nominal capture trajectory. A bounded, force-driven reference adjustment then compensates residual error around this registered trajectory. We evaluate target estimation before capture and docking accuracy after capture separately.

III. PROPOSED METHOD

The method has two stages. Physical-direction-assisted contact-field registration (PCF) first fits a contact boundary to offline poses and directional cues, then aligns online contact poses to this boundary. Registration-anchored capture (RAC) uses the resulting target for nominal motion and lets contact feedback make bounded corrections. The registration sensitivity determines how strongly each correction direction is anchored to the estimate.

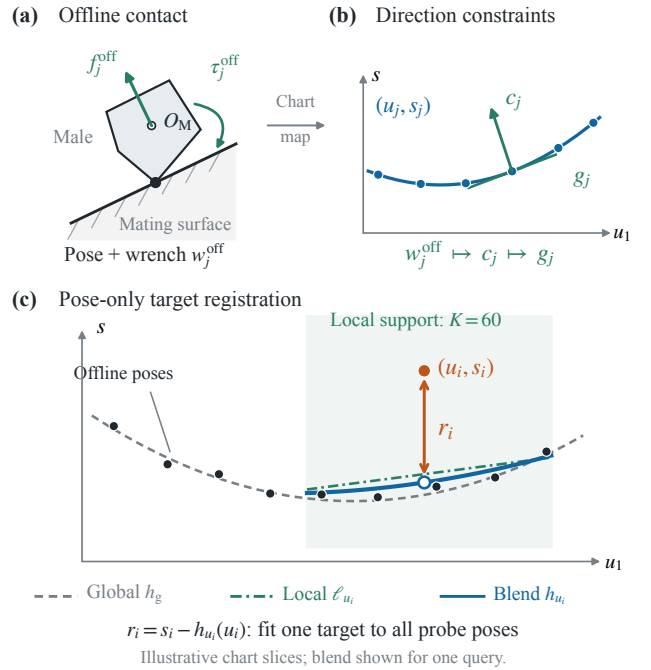


Fig. 2. Physical-direction-assisted contact-field registration. (a) A contact wrench about the male origin supplies a physical direction cue. (b) Mapping a wrench to the pose chart yields a covector and a field-slope constraint. (c) A global field and local affine refinement define the height residual of an online pose. All probe poses estimate one shared target. Curves are illustrative chart slices; the local fit and blend are shown for one query. The coordinate s is not physical insertion depth.

A. Physical-Direction-Assisted Contact-Field Registration

A contact pose tells us where the configuration boundary lies, but sparse poses may not determine its local orientation. A contact wrench provides additional directional information. PCF uses this information to constrain the offline field, rather than requiring online normals for registration [1]. Figure 2 illustrates the construction and its use.

1) *From contact wrench to field slope:* We first express relative poses in a common six-dimensional chart. For a pose X , define

$$\begin{aligned} y(X) &= \text{Log}(T_{\text{ref}}^{-1}X)^\vee, & x(X) &= S_{\mathcal{M}}^{-1}y(X), \\ s &= x_3, & u &= (x_1, x_2, x_4, x_5, x_6)^\top. \end{aligned} \quad (3)$$

Here T_{ref} is the fixed reference pose of the offline chart, and $S_{\mathcal{M}}$ is a positive diagonal matrix of coordinate scales. Neither is an estimated target. The operators $\wedge$ and $\vee$ convert between a twist vector, ordered as translation then rotation, and its Lie-algebra matrix. The standardized third coordinate s is a logarithm coordinate, not physical insertion depth. Within the sampled region, we represent the contact boundary as a height field $s = h(u)$. This local representation need not describe the entire interface configuration space.

Let $w_j^{\text{off}} = [f_j^\top, \tau_j^\top]^\top$ be an offline wrench, expressed in the male frame and taken about its origin. The body twist associated with a chart displacement dx is $d\xi_b = J_r^{\text{SE}(3)}(y_j)S_{\mathcal{M}}dx$, where $J_r^{\text{SE}(3)}$ is the right Jacobian of $\text{SE}(3)$. Preserving virtual work, $(w_j^{\text{off}})^\top d\xi_b = c_j^\top dx$, gives

$$c_j = S_{\mathcal{M}}^\top J_r^{\text{SE}(3)}(y_j)^\top w_j^{\text{off}}, \quad g_j = -\frac{c_{j,u}}{c_{j,s}}. \quad (4)$$

The covector c_j is the contact direction in chart coordinates; $c_{j,s}$ selects its third component and $c_{j,u}$ the remaining five. For a locally smooth, approximately frictionless single contact, it is proportional to the normal of $s - h(u) = 0$. Thus g_j estimates the slope $\nabla h(u_j)$, with the unknown contact-load magnitude canceled by the ratio. This also explains why the wrench must be expressed at the correct origin before the chart mapping.

Friction and multiple contacts make measured aggregate wrenches only approximate normals. We therefore use g_j as a soft derivative observation and discard derivatives with a near-zero height component. Let $\mathcal{D}$ index the usable derivatives. A sample without a usable derivative still contributes its contact position (u_j, s_j) .

2) *Fitting the contact field:* A global field supplies support across the sampled region, while a local fit adapts to nearby contact geometry. Both fits use the same two forms of evidence: contact positions constrain field values, and physical directions constrain field slopes.

For the global field, use $h_g(u) = \psi(u)^\top \theta$, where ψ contains affine and radial basis functions. Its coefficients solve

$$\begin{aligned} \min_{\theta} \sum_{j=1}^N (h_g(u_j) - s_j)^2 + \theta^\top \Gamma \theta \\ + \alpha_g \frac{N}{|\mathcal{D}|} \sum_{j \in \mathcal{D}} \|\nabla h_g(u_j) - g_j\|^2. \end{aligned} \quad (5)$$

The first term fits the observed boundary, the second regularizes the basis coefficients with $\Gamma \succ 0$, and the third enforces the physical slopes. The factor $N/|\mathcal{D}|$ balances the two sample counts; α_g controls their relative weight. When no derivatives are available, the third term is omitted.

At a query $\bar{u}$, let $\mathcal{N}(\bar{u})$ be its K nearest offline input coordinates and let $\mathcal{D}_{\bar{u}} = \mathcal{N}(\bar{u}) \cap \mathcal{D}$. The local field $\ell_{\bar{u}}(u) = a + b^\top(u - \bar{u})$ solves

$$\begin{aligned} \min_{a,b} \sum_{j \in \mathcal{N}(\bar{u})} (a + b^\top(u_j - \bar{u}) - s_j)^2 \\ + \alpha_l \frac{K}{|\mathcal{D}_{\bar{u}}|} \sum_{j \in \mathcal{D}_{\bar{u}}} \|b - g_j\|^2, \quad \alpha_l > 0. \end{aligned} \quad (6)$$

The derivative term directly constrains the local slope b . With at least one usable derivative, this quadratic has a unique solution even if the positions alone do not span all input directions: zero curvature would require a zero slope increment and then a zero intercept increment. This is a property of the field fit, not a claim of target-pose observability.

The field used at the query is

$$h_{\bar{u}}(u) = (1 - \beta)h_g(u) + \beta\ell_{\bar{u}}(u), \quad 0 \leq \beta \leq 1. \quad (7)$$

The blend weight β is fixed before evaluation. If the neighborhood has no usable derivative, the query falls back to h_g . The global fit is computed offline; only the affine refinement is fitted at query time.

3) *Estimating the target from online poses:* Changing the target estimate changes every relative observation in (2). Let (u_i, s_i) be its chart components. We choose the target that brings these observations onto the field by minimizing their signed height residuals:

$$\begin{aligned} r_i(\hat{T}) &= s_i - h_{u_i}(u_i), \\ \Phi(\hat{T}) &= \frac{1}{N_o} \sum_{i=1}^{N_o} \rho_\kappa(r_i(\hat{T})). \end{aligned} \quad (8)$$

Here ρ_κ is the Huber loss with transition κ . All valid probe poses contribute to one shared target estimate; online wrench values are not additional registration residuals.

Starting from $\hat{T}_v$, a damped Gauss-Newton solver uses right increments

$$\hat{T}^+ = \hat{T} \text{Exp}((E_5 \delta)^\wedge), \quad E_5 \delta = (\delta_x, \delta_y, 0, \delta_\phi^\top)^\top, \quad (9)$$

where $\delta \in \mathbb{R}^5$ contains the lateral and rotational increments. A local linearization holds the neighborhoods, fitted coefficients, and robust weights fixed. Each trial target then reselects neighbors and refits the local field; it is accepted only if the recomputed Φ decreases. This accommodates changing local models without claiming differentiability through neighbor selection or global convergence.

The raw estimate $\hat{T}$ is converted into a bounded correction relative to the prior to obtain $\hat{T}_r$. Axial translation in that prior-relative tangent is set to zero, since composition of the local steps can otherwise introduce axial coupling. Correction limits can change the optimizer's result, so the sensitivity used for capture is recomputed at the adopted target $\hat{T}_r$.

B. Registration-Anchored Constrained Capture

RAC compensates residual registration error during insertion. Contact can move the reference, while an elastic anchor discourages unnecessary departure from the registered trajectory.

1) *Correction coordinates and contact effort*: In the adopted target frame, let $(p_n(t), R_n(t))$ be the nominal male reference. Write its correction as $q = [p_c^\top, \phi^\top]^\top \in \mathbb{R}^5$, where $p_c \in \mathbb{R}^2$ is a lateral translation and $\phi \in \mathbb{R}^3$ a right rotation vector. With $E = [e_1 \ e_2]$ formed from the first two standard basis vectors, the corrected reference is

$$p_r(t, q) = p_n(t) + E p_c, \quad R_r(t, q) = R_n(t) \text{Exp}([\phi]_\times). \quad (10)$$

Thus p_c is expressed in target-frame metres, whereas ϕ is expressed in the nominal reference's body coordinates in radians. The skew matrix $[\phi]_\times$ represents the cross product with ϕ . Initialized at $q_0 = 0$, the reference initially follows the registered nominal trajectory. Axial motion is regulated separately and is not a decision variable in q .

At a fixed nominal time, a correction velocity $\dot{q}$ induces the body twist $V_c = B\dot{q}$. Its power-conjugate effort is

$$B(q, t) = \begin{bmatrix} R_r^\top E & 0 \\ 0 & J_r^{\text{SO}(3)}(\phi) \end{bmatrix}, \quad Q = B^\top w_r. \quad (11)$$

Here $J_r^{\text{SO}(3)}$ is the right Jacobian of $\text{SO}(3)$, and w_r is the contact wrench about the reference origin, expressed in the reference frame. This mapping ensures $Q^\top \dot{q} = w_r^\top V_c$ and requires the full wrench, including the moment-arm transformation from the sensor origin.

2) *Transferring registration sensitivity to the anchor*: A reference correction should be resisted more strongly along directions well constrained by the probe. To quantify this directional sensitivity, linearize the registration residuals at $\hat{T}_r$ using a full six-dimensional right perturbation ξ :

$$J_i = \left. \frac{\partial r_i(\hat{T}_r \text{Exp}(\xi^\wedge))}{\partial \xi} \right|_{\xi=0}, \quad (12)$$

$$\mathcal{H}_T = \sum_i \omega_i J_i^\top J_i, \quad \omega_i \geq 0, \quad \sum_i \omega_i = 1.$$

The query neighborhoods, local coefficients, and normalized Huber weights ω_i are frozen at this target. The matrix $\mathcal{H}_T$ measures local residual sensitivity; it is neither a calibrated covariance nor a contact-stiffness estimate.

The target perturbation ξ and the reference correction q use different coordinates. Let X_g be the nominal terminal male pose relative to the adopted target, and evaluate B_g there at $q = 0$. To first order, the target perturbation equivalent to a tool correction dq is $\xi = A_c dq$, giving

$$A_c = -\text{Ad}_{X_g} B_g, \quad \mathcal{H}_q = A_c^\top \mathcal{H}_T A_c. \quad (13)$$

The adjoint Ad_{X_g} maps twist coordinates between these frames and retains the translation-rotation coupling due to their offset origins. Computing the full $\mathcal{H}_T$ before this transfer is therefore necessary even though capture has only five correction variables.

Let $D \succ 0$ be a diagonal mobility that sets the relative translation and rotation response. We define an anchor stiffness Λ and its virtual storage U_a by

$$G = D^{1/2} \mathcal{H}_q D^{1/2}, \quad A = \varepsilon I_5 + (1 - \varepsilon) \frac{5G}{\text{tr} G}, \quad (14)$$

$$\Lambda = \tau_a^{-1} D^{-1/2} A D^{-1/2}, \quad U_a(q) = \frac{1}{2} q^\top \Lambda q.$$

For $\text{tr} G > 0$ and $0 < \varepsilon \leq 1$, the shape matrix satisfies $\text{tr} A = 5$ and $A \succeq \varepsilon I_5$. Consequently, the time constant $\tau_a > 0$ sets the overall restoring scale, while A redistributes it according to registration sensitivity. The floor ε retains an anchor in weak directions. The entries of D have units $q_i^2/(\text{Js})$, so U_a has units of virtual energy. Both Λ and its reference $q = 0$ remain fixed during capture. The storage is a designed regularizer, not the exact registration loss.

3) *Balancing contact response and bounded correction*:

At step k , let $\Delta t_k > 0$ be the update interval, q_k the current correction, and Q_k the measured generalized effort. The next candidate correction solves

$$q_k^* = \arg \min_{q^+ \in \mathcal{C}_k} \frac{\|q^+ - q_k\|_{D^{-1}}^2}{2\Delta t_k} + U_a(q^+) - Q_k^\top (q^+ - q_k). \quad (15)$$

where $\|v\|_W^2 = v^\top W v$. The first term penalizes rapid reference changes, the second penalizes departure from registration, and the last allows contact effort to drive the correction. This proximal update [10] is an implicit anchored admittance step when no constraint is active.

The feasible set bounds both the cumulative correction and its command rate. With positive componentwise limits $\bar{q}$ and $v_{\max}$, it is

$$\mathcal{C}_k = \{q^+ : -\bar{q} \leq q^+ \leq \bar{q},$$

$$-v_{\max} \leq M_k(q^+ - q_k)/\Delta t_k \leq v_{\max}\}, \quad (16)$$

$$M_k = \text{Diag}((E^\top R_{r,k} E)^{-1}, J_r^{\text{SO}(3)}(\phi_k)).$$

The lateral projection is assumed nonsingular in the operating region. The matrix M_k converts a correction increment into the five channels used by the reference generator; it differs from the six-dimensional twist map B_k . These limits apply to the correction commands, not to nominal insertion or separate axial regulation. Since a zero increment is feasible, $q_k \in \mathcal{C}_k$ whenever q_k satisfies the cumulative bounds.

The rotational rate constraint is linearized. Before applying the candidate, we also check the finite correction rotation on $\text{SO}(3)$, with the nominal pose held fixed. If needed, the entire five-dimensional step is reduced by one common factor $\alpha \in [0, 1]$: $q_{k+1} = q_k + \alpha(q_k^* - q_k)$. Common reduction preserves convex feasibility and the storage property below. Setting $\Lambda = 0$ removes the anchor without changing the correction coordinates or constraints.

4) *Storage property of the reference update*: **Proposition 1.** *Let $D \succ 0$, fixed $\Lambda \succeq 0$, and $\Delta t_k > 0$. If $\mathcal{C}_k$ is closed and convex and contains q_k , (15) has a unique solution. For the implemented increment $d_k = \alpha(q_k^* - q_k)$ with $0 \leq \alpha \leq 1$,*

$$U_a(q_{k+1}) - U_a(q_k) \leq Q_k^\top d_k - \frac{\|d_k\|_{D^{-1}}^2}{\Delta t_k} - \frac{1}{2} \|d_k\|_{\Lambda}^2. \quad (17)$$

Proof: The objective is coercive and strictly convex because its Hessian is $D^{-1}/\Delta t_k + \Lambda \succ 0$. Write $d^* = q_k^* - q_k$. The optimality condition, evaluated at the feasible point q_k , gives

$$Q_k^\top d^* \geq \|d^*\|_{D^{-1}}^2 / \Delta t_k + (d^*)^\top \Lambda q_k^*.$$

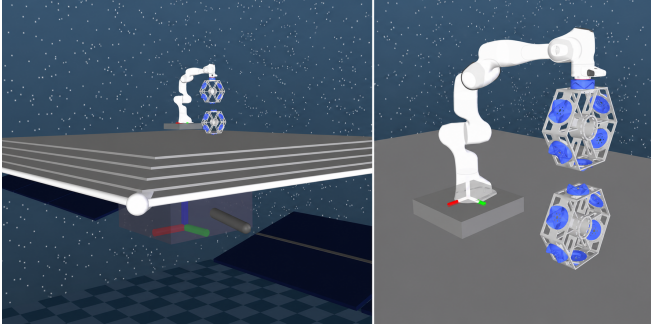


Fig. 3. The MuJoCo-based simulation environment, showing the Franka Emika Panda manipulator approaching the lotus-shaped interface.

Expanding the quadratic storage proves (17) for $\alpha = 1$. For a common step reduction, the slack equals α times the full-step slack plus $\alpha(1 - \alpha)(\|d^*\|_{D^{-1}}^2 / \Delta t_k + \|d^*\|_{\Lambda}^2) \geq 0$. Convexity preserves feasibility. $\square$

The inequality bounds the virtual storage increase by sampled contact work minus nonnegative motion penalties, including common step reduction. It does not establish passivity of the whole robot or guarantee assembly completion: nominal insertion, axial regulation, filtering, and tracking dynamics lie outside this subsystem [11]. Capture accuracy and the effect of the anchor are evaluated experimentally.

IV. EXPERIMENTAL EVALUATION

A. Simulation Analysis

Figures 3–5 show the MuJoCo task, interface geometry, and 50,000-pose offline reference. Registration uses the full configurations. Physical derivative labels in simulation are obtained from the deepest contact in the collision geometry.

1) *Setup and fixed parameters*: Both platforms use the field construction of Section III-A, with $K = 60$, $\alpha_g = \alpha_l = 0.1$, and Huber transition $\kappa = 0.1$. The global basis uses six affine and 384 thin-plate radial terms. The affine ridge weight is 10^{-8} ; radial width/ridge weight are $1/0.01$ in simulation and $4/1$ on hardware. The respective blend weights are $\beta = 0.84714$ and 0.23797 . Coordinate scales use offline standard deviations with positive floors. Derivative labels require finite slopes and $|c_{j,s}|/\|c_j\| \geq 0.1$, with an additional load gate on hardware. Settings were fixed after development on offline acquisition blocks; coefficients are refitted on all offline samples, without a held-out field test.

RAC uses $\varepsilon = 0.2$ and $\tau_a = 5$ s on both platforms. Capture uses the full filtered wrench after per-trial calibration and frame/origin conversion, before componentwise dead zones or clipping. Simulation uses torque-level impedance. Hardware follows the reference through inverse kinematics and a position servo; its downstream admittance block receives zero external wrench.

We evaluate registration separately from docking. The four systems are nominal compliant capture (Compliance), PCF followed by that capture (PCF), RAC without target registration (RAC only), and PCF+RAC. Compliance starts at contact; the other three share a 10-s probe and capture

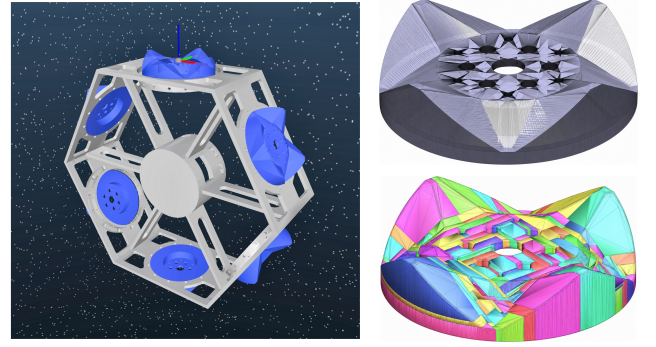


Fig. 4. Docking module and interface models: the module to be assembled (left), the interface mesh (upper right), and its convex decomposition using CoACD (lower right).

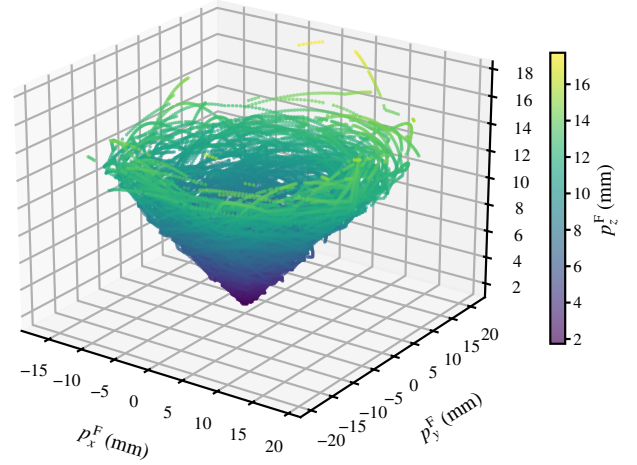


Fig. 5. Position projection of the 50,000-pose simulation contact set, colored by female-frame axial position p_z^F . Registration uses the complete six-dimensional samples.

initial physical and filter state. RAC only retains the visual prior and computes sensitivity there; PCF+RAC computes it at the adopted target. Both use Section III-B's update and constraints. The two PCF branches share their adopted estimate.

All systems share the execution layer, nominal motion, and axial policy. Capture has 10-s execution (5-s nominal motion) and a 2-s fixed-reference hold. Completion requires both 3D position and geodesic rotation RMS over $[10, 12]$ s to be at most 0.5 mm/ 0.5° ; squared errors are integrated and divided by exactly 2 s. Capture uses the fixed seated-interface datum only for scoring, distinct from the female CAD registration datum. The model has no locking stage. Capture correction bounds are ± 10 mm per lateral and $\pm 2^\circ$ per rotation-vector component, with rates 1 mm/s and $0.5^\circ/\text{s}$ and $D = \text{Diag}(0.05, 0.05, 5, 5, 5)$. The prior-relative registration increment has zero axial tangent translation, a 12-mm translation-norm limit, and rotation-vector clipping at $\pm 2^\circ$ per component, rather than a total-angle limit. Raw

TABLE I
TARGET-FRAME BOUNDARY BIAS MATRIX.

Case	$[\Delta x, \Delta y]$ [mm]	$[\Delta \phi_x, \Delta \phi_y, \Delta \phi_z]$ [deg]
B00	[0, 0]	[0, 0, 0]
B01/B02	$[\pm 10, 0]$	[0, 0, 0]
B03/B04	$[0, \pm 10]$	[0, 0, 0]
B05/B06	[0, 0]	$[\pm 2, \pm 2, 0]$
B07/B08	[0, 0]	$[\pm 2, 0, \pm 2]$
B09/B10	[0, 0]	$[0, \pm 2, \pm 2]$
B11/B12	$[\pm 6, \mp 6]$	$[\pm 2, \mp 2, \pm 2]$

TABLE II
PAIRED SIMULATION REGISTRATION WITH 50K OFFLINE CONTACTS.

Method ($K = 60$)	e_{xy} [mm]	e_R [deg]	e_N	Time [s]
Symmetric CMR	1.478	1.984	4.246	3.409
PCF (ours)	0.428	0.469	1.047	1.956

registration is scored before these limits.

Table I gives physical translation and rotation-vector components, with zero axial translation; paired signs follow case order. Two-axis and three-axis rotation magnitudes are 2.828° and 3.464° . Chosen during development, these boundary conditions and algorithm parameters are fixed for ten exploration groups. They are separate from camera-conjugated errors. Variation measures exploration sensitivity, not random-task success probability; Compliance is deterministic.

2) *Paired registration accuracy*: Each estimator receives identical online observations and prior, without executing capture. All use $K = 60$; CMR queries six-dimensional configuration coordinates, whereas PCF queries five-dimensional input coordinates. Symmetric CMR uses offline and online PCA normals with at most 100 iterations; PCF uses at most 20 iterations. This compares complete estimators with their frozen solver settings, not only one changed residual.

Table II averages all 36 nonzero inputs from seeds 0–2 of Table I (452–604 contacts); corresponding PCF estimates are reused in capture. Raw errors precede execution limits: e_{xy} is lateral distance, e_R is geodesic rotation error, and $e_N = \max(e_{xy}/(0.5 \text{ mm}), e_R/0.5^\circ)$ is computed per input, then averaged. Timing uses a warm sequential pass on one pinned i7-14650HX CPU, single-thread linear algebra, and alternating method order. It includes solver work and neighbor queries, excluding offline loading, probing, scoring, and file I/O, with no concurrent simulation.

PCF reduces mean lateral/rotation error by 71.0%/76.4% relative to single-start Symmetric CMR. Joint tolerance is attained on 23/36 inputs, versus 0/36 for Symmetric CMR. These gains are specific to the tested structured target matrix.

3) *Four-system completion and representative capture*: Table III retains all 520 runs from seeds 0–9, varying exploration phases, amplitudes, and center offsets at fixed biases. Cells count passes out of ten; paired cases follow case order. The last rows give per-group means and sample SDs. Deterministic Compliance repeats are not independent evidence. Under 10-mm single-axis translation, PCF, RAC

TABLE III
SIMULATION COMPLETION OVER TEN EXPLORATION GROUPS (COUNTS OUT OF TEN).

Case	Compliance	PCF	RAC only	PCF+RAC
B00	10	10	10	10
B01/B02	0, 0	5, 8	1, 1	10, 10
B03/B04	0, 0	9, 9	1, 0	10, 10
B05/B06	0, 0	9, 4	10, 10	10, 10
B07/B08	0, 0	8, 7	10, 10	10, 10
B09/B10	0, 0	9, 9	10, 4	10, 10
B11/B12	0, 0	8, 7	9, 0	10, 9
Within tolerance	10/130	102/130	76/130	129/130
Mean /13	1.00	10.20	7.60	12.90
Sample SD	0.00	1.03	0.97	0.32

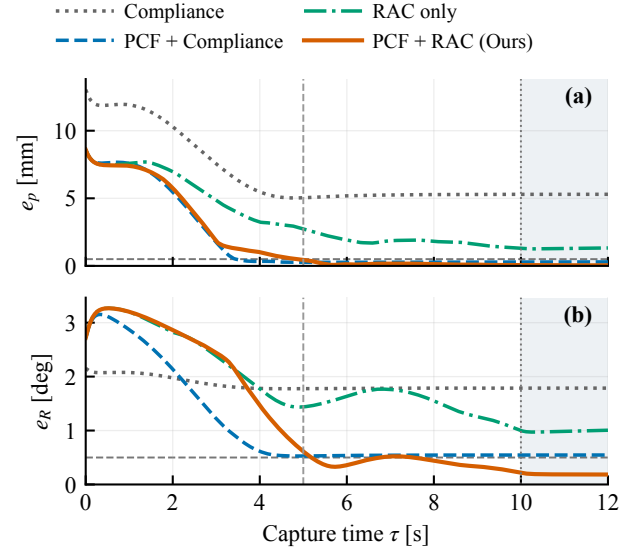


Fig. 6. Four-system camera C11 capture, displaying seed 0 fixed before the runs: (a) three-dimensional position error and (b) geodesic rotation error. Compliance starts at contact; the other systems share a 10-s probe. Nominal motion ends at 5 s and adjustment at 10 s; the shaded 10–12-s hold defines the reported RMS.

only, and PCF+RAC complete 31/40, 3/40, and 40/40 captures. With two-axis rotation they complete 46/60, 54/60, and 60/60; with combined translation and three-axis rotation, 15/20, 9/20, and 19/20. Thus the tested conditions expose complementary limitations of registration followed by compliance and capture at the prior. The sole full-method miss is seed 2, B12: raw registration leaves $3.850 \text{ mm}/4.024^\circ$ lateral/rotation error, and capture reaches a rotation-component bound, ending at $1.255 \text{ mm}/2.005^\circ$ hold RMS.

Mean position/rotation hold errors for the four systems are $3.279 \text{ mm}/1.953^\circ$, $0.282 \text{ mm}/0.432^\circ$, $0.389 \text{ mm}/0.373^\circ$, and $0.117 \text{ mm}/0.180^\circ$, respectively. A paired application-limit check retains the earlier total-angle 2° cap on the same raw estimates: PCF then completes 41/130 and PCF+RAC 129/130. Table III uses the stronger componentwise PCF baseline, separating estimator error from execution clipping.

Figure 6 uses a separate camera-frame bias: $[2, -2, 0]$

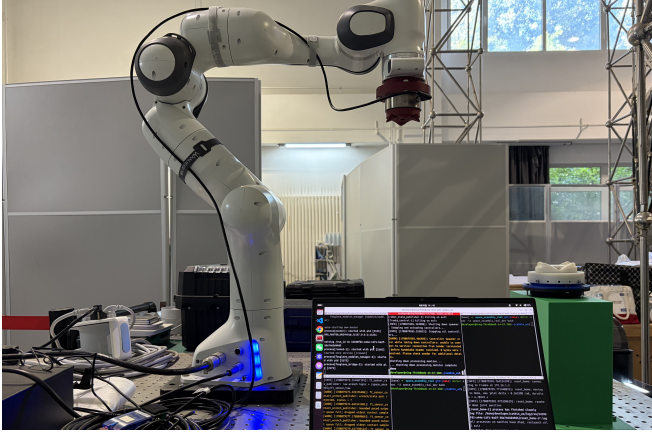


Fig. 7. Ground-based docking platform. The male interface is mounted through a six-axis force/torque sensor and approaches the fixed female interface.

mm translation and xyz RPY $[0.5, -0.5, 2]^\circ$, with $\hat{T}_v = T_C \Delta T_C T_C^{-1} T_F$, where $T_C = {}^B T_C$ is the camera pose and ΔT_C is the camera-frame error. Before probing, its target-pose error is 10.900 mm/2.123°. Across the same ten exploration seeds, the four camera completion counts are 0/10, 5/10, 1/10, 9/10; corresponding mean hold errors are 5.296 mm/1.786°, 0.581 mm/0.741°, 1.140 mm/0.932°, 0.105 mm/0.213°. The PCF branches use Table III’s componentwise execution rule. This case does not establish a capture range.

A paired PCF+RAC control replaces the sensitivity shape with $A = I_5$ while retaining the registered target, constraints, time constant, and capture initial state. On the current boundary matrix it completes 127/130, with mean hold errors 0.116 mm/0.200°, versus 129/130 and 0.117 mm/0.180° for the full shape. In the camera condition, the isotropic and full shapes complete 10/10 and 9/10. Earlier controls on 14 frictionless and four synthetic $\mu = 0.4$ conditions gave 18/18 for both information and same-spectrum rotated anchors; the rotated anchor had slightly lower mean errors. Unanchored and isotropic controls gave 17/18. These controls do not establish a unique advantage of the information directions; the synthetic friction is not a measured value.

B. Experimental Studies

Figures 7 and 8 show the hardware platform and its 19,843-pose same-installation contact set. Its physical derivative labels use aggregate measured wrench, unlike the geometric labels in simulation. Transformations use the male-interface frame and include the sensor-to-interface moment arm. The calibrated alignment defines the offline datum and scoring reference; online solvers receive the biased prior and contact observations. This reference is not independently metrologized.

1) *Paired registration replay*: The available September 9 records comprise B00–B12 and two camera trials, with 20-s hardware probes rather than simulation’s 10 s. They participated in development; this is not prospective frozen-method validation. Table IV uses their 12 nonzero target

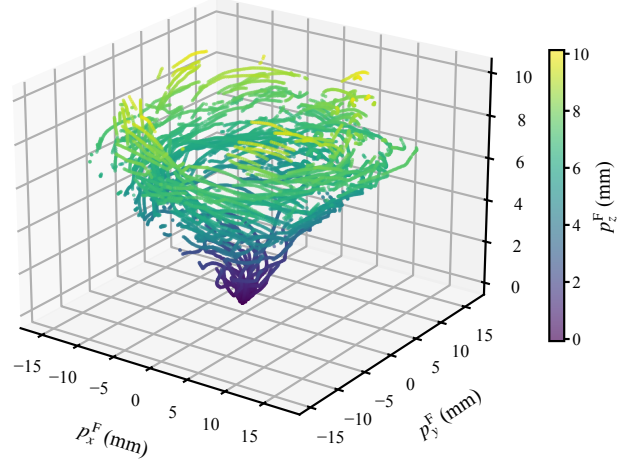


Fig. 8. Position projection of the 19,843-pose same-installation hardware contact set, expressed in the calibrated aligned-male target datum $\{F\}$. Positions are recovered from full relative transforms; registration retains the six-dimensional configurations.

TABLE IV
PAIRED HARDWARE REGISTRATION REPLAY, $K = 60$.

Method	e_{xy} [mm]	e_R [deg]	e_N	Time [s]
Symmetric, one start	1.790	1.201	3.706	4.689
Symmetric, five starts	1.619	1.114	3.321	24.400
PCF (ours)	0.897	0.619	1.853	3.262

cases and the same current offline reference and priors for all estimators. These historical hardware case labels refer to their recorded biases, not the new simulation boundary matrix in Table I.

The table reports arithmetic means. The task-adapted five-start Symmetric baseline averages estimates on SE(3) from the prior and four Gaussian right perturbations ($\sigma = [2, 2, 0]$ mm and $[1, 1, 1]^\circ$, seed 202609091), without truth-based selection. Of its 60 starts, 48 reach the 100-iteration cap. Timing follows the sequential CPU protocol of Table II; five-start time includes all solves and pose averaging.

PCF reduces lateral/rotation means by 44.6%/44.4% relative to the five-start baseline; 10/12 cases improve in both metrics, but only 2/12 meet joint tolerance, versus 0/12 for that baseline. PCF perturbs B00 to 0.745 mm/0.199°. The two camera records use 1° yaw, unlike simulation C11; PCF averages 1.412 mm/0.791° versus 2.084 mm/1.613° for the five-start baseline.

A matched direction ablation retains all 19,843 positions and the same 8,769 valid derivative labels. Physical versus PCA directions yield 0.941 mm/0.650° versus 1.310 mm/1.046° on these 12 records. Removing both derivative terms instead gives 0.761 mm/1.028°: translation can improve while rotation worsens. These controls support useful physical information without claiming that every term is indispensable.

TABLE V
HARDWARE TARGET-FRAME COMPLETION: NEW BATCH PENDING.

Case	Compliance	PCF	RAC only	PCF+RAC
B00	–	–	–	–
B01/B02	–	–	–	–
B03/B04	–	–	–	–
B05/B06	–	–	–	–
B07/B08	–	–	–	–
B09/B10	–	–	–	–
B11/B12	–	–	–	–
Within tolerance	–	–	–	–

New-method hardware traces pending
Target B11: Compliance / PCF / RAC only / PCF+RAC

Fig. 9. Planned hardware target-frame B11 response comparison using position and geodesic rotation errors. New-method measurements are pending.

2) *Four-system docking and camera-bias range:* The planned hardware batch uses the same four system roles: Compliance starts at contact; RAC only, PCF, and PCF+RAC share a 20-s probe. RAC only retains the prior; the registered branches use the PCF target. Capture uses 10-s execution and 2-s hold, with [10,12]-s RMS and $D = \text{Diag}(0.04, 0.04, 5, 5, 5)$. Only software integration checks of PCF+RAC are complete; new hardware capture measurements remain pending.

Table V reserves the B00–B12 matrix. Dashes and the outlined panels denote pending new-method hardware measurements.

Figures 9 and 10 reserve the hardware transient and camera-bias-range tests. Record the base camera bias and extrinsic before execution; recompute target displacement after conjugation. Repeated measurements are required to establish a capture boundary.

V. CONCLUSION

We presented contact-field registration followed by anchored capture. Physical direction cues constrain field slopes and improve the tested target estimates. A bounded reference optimization accommodates residual error with a discrete storage inequality. Across ten exploration groups, PCF+RAC completes 129/130 target-frame boundary captures, versus 102/130 for registered compliance and 76/130 for capture at the prior, and 9/10 camera-case captures. The retained failure limits the demonstrated range. Hardware registration error remains limiting, and a unique benefit of the anchor directions is not established. Prospective hardware capture and range tests remain necessary. The theory concerns the reference subsystem, not whole-robot stability or guaranteed docking completion.

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New-method camera-bias sweep pending
Prescribed scales: 1.00, 1.25, 1.50, 1.75, 2.00

Fig. 10. Planned PCF+RAC hardware camera-bias sweep. Each scale will report initial target error and two-second hold RMS. Measurements are pending.

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